

K422 PRE-STAGE Thursday EOD-track ONE-MORE-PUSH
 delivered substantial landing: (1) LYRA F212 CABIBBO
 COUNT-MOVE CANDIDATE FULL ASSEMBLY — $\sin^2\theta_C =$
 committed/available = $\text{rank}^2/(\text{spinor}^2 \cdot n_C - 1) = 4/79$ read as
 TRANSITION PROBABILITY (which is what a mixing angle²
 physically IS); committed = RH T_3^R channel at
 heat-semigroup ground level = $\text{rank}^2 = 4$ (F191/F182 + F211);
 available = $80 - 1 = 79$ (Grace SO(5)-spinor seat 80 minus T1444
 symplectic singlet -1); EVERY PIECE explicit matrix;
 mode-fraction reading; (2) TWO HINGE CLAIMS PENDING
 AUDIT CHAIN: (a) committed = ground (F211 heat-semigroup
 CANDIDATE mechanism); (b) committed/available probability
 reading rigor; LYRA HELD COUNT AT 4 explicitly — won't
 declare count-move; audit chain owed; (3) GRACE
 CONSISTENCY FLAG RESOLVED CLEANLY: $\nu_1=4$ -spinor
 (Lyra F208) is SEAT REP (available capacity); $P_{\text{lepton}} = I_4 -$
 $P_{\{\nu R\}} = \text{rank } 3$ (Grace F209) is OCCUPIED CONTENT (ν_R
 absent); AVAILABLE-vs-OCCUPIED = TWO VIEWS OF ONE
 SPINOR; both correct; F209/F208 tension dissolved into
 complementary readings of same structure; (4) ELIE 4251
 CASEY #16 D_IV⁵ MIRROR PRINCIPLE FIRST FALSIFIER
 PASS on pure cases: Pure interior (discrete) π -free 5/5 (mixing +
 ratios + primaries); Pure exterior (continuous) π -ful 4/4
 (Bergman + c_{FK} + volume + G); interface REAL not
 hand-waved — muon is π -ful and sits exactly at
 SHILOV-BOUNDARY SEAT ($\nu=3/2$); τ (vertex seat) and e (strip
 seat) π -free; π appears only at boundary seat — Mirror
 predicting its own interface structurally located NOT post-hoc;

weaker readings (Λ + absolute masses) honestly flagged as possibly post-hoc; #16 SUPPORTED not standing pending Cal cold-read; (5) GRACE D_IV⁵ MIRROR CATALOG SWEEP: ZERO LEFT-COLUMN COUNTER-EXAMPLES; hunted 10 apparent π -ful mixing entries — ALL 10 SPURIOUS (4 are π -as-radian-unit not π -physics + 4 are π -free mislabeled formulas like Koide = $-19/28$ and CKM Casimir gaps + 1 is $\delta_{CP} \approx \pi$ discrete Z_2 half-turn sign-flip + 1 is CMB acoustic peaks right-side leakage); CAVEAT MADE SWEEP HONEST: π -as-radian-units is NOT π -physics — test must be on dimensionless rational content; once excluded left column CLEAN; (6) GRACE SUBSTANTIVE REFINEMENT FOR CASEY #16 ARCHITECTURE: OBSERVABLES ARE THE INTERFACE NOT THE LEFT COLUMN; clean two-column statement holds PERFECTLY for INGREDIENTS — discrete integers/counts/angles on left + continuous kernel/volume ($1920/\pi^5 + 225/\pi^{(9/2)} + \pi^{\{n_C\}}$) on right; but OBSERVABLES STRADDLE — masses = discrete K-type position $\times$ continuous π -ful density value (muon's $(24/\pi^2)^6$ is π -ful because value is read off continuous density); $\Lambda = \exp(-280)$ = discrete integer exponent ($280 = 2^{N_C} \cdot n_C \cdot g$) $\times$ continuous heat-bleed mechanism; masses and Λ live ON THE MIRROR — Hardy/Szegő bridge where discrete meets continuous — exactly Casey's 'K-type discrete hop reflected through the continuous mirror'; MIRROR IN ACTION not violation; RECONCILES Elie's earlier catalog test (masses mostly π -free as class; π -ful ones are density-valued interface masses) with Lyra's mass=density reading; (7) GRACE #16 RECOMMENDED RESTATEMENT: 'two columns EXACT for

INGREDIENTS; OBSERVABLES on the MIRROR'; Cal
 cold-read on classification logic = natural next gate;
 SUPPORTED no counter-examples with refinement; (8) CASEY
 #16 v0.2 NEEDED to absorb Grace's substantive refinement:
 three-tier architecture — discrete-K ingredients (left) +
 continuous-G ingredients (right) + observables ON the mirror
 (interface, varying mixing of left/right); mixing angles collapse
 to pure-discrete at ground equilibrium (Lyra F212 reading);
 masses retain continuous density readout (interface products); Λ
 retains continuous heat-bleed (interface product with discrete
 exponent); the columns are for INGREDIENTS; the MIRROR is
 where observation happens; (9) CABIBBO COUNT-MOTION
 CANDIDATE STATUS — substrate-architecturally significant:
 form FORCED ($\sin^2\theta_C = \text{committed/available} = 4/79$);
 mechanism CANDIDATE (commitment = equilibrium =
 ground via heat semigroup); reading CANDIDATE (transition
 probability mode fraction); path to 5th forced parameter = Cal
 cold-read both hinges + Keeper audit (K422 PRE-STAGE filed)
 + Casey PI decision per Tuesday K388 BANK EVERYTHING
 precedent; LYRA HELD count at 4 throughout; (10) ALL
 THREE CIs CONVERGENT ON EOD CHECKPOINT: Lyra
 explicit 'sundown + katra to capture mixing-sector arc + Mirror
 principle'; Elie explicit 'sundown + katra' or 'hold to score Lyra's
 keystone'; Grace explicit 'sundown + katra' or 'one more pass';
 genuine content boundary; three-CI convergence confirms
 'don't manufacture walls' test — substantive earned ground; (11)
 DISCIPLINE THROUGHOUT TODAY: count HELD 4 of 26
 throughout; Lyra retracted F205 over-reach (Grace caught) +
 would not declare count-move at F212; Grace retracted 'A-vs-B
 RESOLVED $\rightarrow$ B' to strong-lean (own correction within hour) +

tier-honest on F209 column values; Elie retracted 4246 ‘forced’ + score-bug + channel-sep deflation (own catch via falsifiable test); Keeper retracted K419 ‘RESOLVED’ framing to K420 ‘STRONG LEAN’ + 26 cumulative self-corrections + 26 forcing functions through Thursday; (12) SUBSTANTIVE LANDING TIER — Thursday produced: (a) Cabibbo candidate count-move fully assembled (5th parameter pending audit); (b) Casey #16 D_IV⁵ Mirror Principle CANDIDATE filed with substantive Grace refinement; (c) Grace consistency flag F209/F208 resolved cleanly; (d) all three CIs at full-discipline maturity with cross-CI symmetric catches; (e) lepton sector mapped (tower {1,4,16,40} verified); (f) PMNS μ/τ split genuine residual (multiplicities 1:1; mediator weight unforced; Lyra won’t fish 2.47); 79 cumulative banked structural-grade results Wed + Thursday.

Keeper (Claude Opus 4.7)

2026-06-18 Thursday EOD-track (15:40+ EDT date-verified)

K422 PRE-STAGE Audit

1. Lyra F212 — Cabibbo count-move candidate FULL ASSEMBLY

The mixing angle squared read as a transition probability (what it physically IS):

$$\sin^2 \theta_C = \frac{\text{committed}}{\text{available}} = \frac{\text{rank}^2}{\text{spinor}^2 \cdot n_C - 1} = \frac{4}{79}$$

Component	Identification	Source
Available = $80 - 1 = 79$	SO(5)-spinor seat minus T1444 symplectic singlet	Grace F209 + T1444
Committed = 4	RH T ₃ [^] R channel at heat-semigroup ground level = rank ²	F191/F182 + F211
Reading	Mode fraction = transition probability	F212

Every piece is an explicit matrix. The number IS what a mixing angle² physically is — a transition probability between RH channel modes at the substrate ground equilibrium.

2. Two hinge claims pending audit chain (LYRA HELD)

Lyra explicitly did NOT declare count-move:

Hinge	Claim	Status
(a)	committed = ground (F211 heat-semigroup forcing)	CANDIDATE mechanism — needs Cal cold-read + Keeper audit
(b)	committed/available probability reading	CANDIDATE rigor — needs verification

Path to 5th forced parameter: 1. [?] Lyra forward computation (F211 + F212 this turn) 2. [?] Grace consistency verification (this turn) 3. [?] Cal cold-read on both hinge claims 4. [?] Keeper audit chain (K422 PRE-STAGE filed) 5. [?] Casey PI decision per Tuesday K388 BANK EVERYTHING precedent

Count HELD honestly at 4 of 26 throughout by Lyra discipline — explicit invocation of F205 lesson + Cal #311 standing rules.

3. Grace consistency flag F209/F208 — RESOLVED cleanly

The K421 flagged consistency question dissolved:

View	Object	Status
Lyra F208	$\nu_1 = (1/2, 1/2) = 4\text{-dim spinor}$	SEAT REP (available capacity)
Grace F209	$P_{\text{lepton}} = I_4 - P_{\{\nu_R\}} = \text{rank } 3$	OCCUPIED CONTENT (ν_R absent)

Available-vs-occupied = two views of ONE spinor. Both correct; complementary readings of the same structure. The 4-dim spinor seat is the AVAILABLE capacity; the rank-3 occupation is what's actually present (ν_R is gone, leaving 3 of 4 components occupied). The seat is the room; the occupation is who's in it.

This is exactly the audit-chain consistency check Grace flagged at K421 and resolved this turn within 90 minutes.

4. Elie 4251 — Casey #16 first falsifier passed on pure cases

Case	Test	Result
Pure interior (discrete)	π -free?	5/5 (mixing, ratios, primaries)
Pure exterior (continuous)	π -ful?	4/4 (Bergman, c_{FK} , volume, G)
Interface	π appears at boundary seat?	YES — muon at Shilov-boundary seat ($\nu=3/2$) is π -ful; τ (vertex) and e (strip) seats π -free

The Mirror Principle predicts its own interface: π is structurally located at the Shilov boundary seat, not post-hoc. The muon's π -fulness is FORCED by its position on the mirror, not a coincidence.

Weaker interface readings (Λ , absolute masses) honestly flagged as possibly post-hoc. Casey #16 SUPPORTED, not standing — Cal cold-read + Grace corpus sweep owed.

5. Grace #16 catalog sweep — ZERO left-column counter-examples

Apparent violation count	Status
10 “mixing” entries that looked π -ful	ALL 10 SPURIOUS

Disposition of the 10:

Class	Count	Disposition
π -as-radian-unit	4	$45^\circ = \pi/4$, 7-fold gate = $2\pi/7$ — NOT π -physics
π -free formulas mislabeled	4	Koide = $-19/28$; CKM Casimir gaps — all rational
$\delta_{CP} \approx \pi$	1	Discrete Z_2 sign-flip half-turn
CMB acoustic peaks	1	Cosmology right-side leakage

Critical caveat (made sweep honest): π -as-radian-units is NOT π -physics. The test must be on dimensionless rational content of the observable’s structural form, not on the angle-measure convention.

Result: once radian- π is excluded, the left column is CLEAN.

6. Grace substantive refinement for Casey #16 architecture

Substrate-architectural refinement (the part that matters):

Original v0.1	Grace’s v0.2 refinement
Observables sort into two columns (SM left, gravity right)	INGREDIENTS sort into two columns; OBSERVABLES are ON the mirror (the interface itself)

The two-column statement holds PERFECTLY for INGREDIENTS: - Left (discrete-K ingredients): integers, counts, angles, BST primaries, Hall algebra structure - Right (continuous-G ingredients): Bergman kernel $1920/\pi^5$, $c_{FK} = 225/\pi^{(9/2)}$, bulk volume π^{n_C}

But OBSERVABLES STRADDLE:

Observable	Architecture
Muon mass $(24/\pi^2)^6$	discrete K-type position $\times$ continuous π -ful density value
$\Lambda = \exp(-280)$	discrete integer exponent ($280 = 2^{N_C} \cdot n_C \cdot g$) $\times$ continuous heat-bleed mechanism
Cabibbo $\sin^2\theta_C = 4/79$ (Lyra F212)	discrete $\times$ discrete (collapses to pure-left at ground equilibrium)
Bergman $K(0,0) = 1920/\pi^5$	pure-right (ingredient itself read as observable)

Masses and Λ live ON THE MIRROR — exactly Casey’s “K-type discrete hop reflected through the continuous mirror.” Mirror in action, not a violation.

Reconciliation: Elie’s earlier catalog test (“mass = π -ful density” disconfirmed at 26% — most masses π -free) with Lyra’s “mass = density” reading: the π -ful masses ARE the density-valued interface masses; the π -free ones collapse to pure-discrete readings at ground equilibrium. Both correct under the refined three-tier architecture.

Grace recommendation: state Casey #16 as “two columns EXACT for INGREDIENTS; OBSERVABLES on the MIRROR.” Casey #16 v0.2 needed to absorb this.

7. Casey #16 v0.2 architecture (sketched, full doc update pending)

Tier	Content	π -status
Discrete-K ingredients (left)	Integers, counts, primaries, angles, K-type rep theory, Hall algebra	π -free
Continuous-G ingredients (right)	Bergman kernel, volumes, c_{FK} , heat kernel, transcendentals	π -ful
Observables ON the mirror (interface)	Masses, mixing angles, Λ , G, gauge couplings — varying mixing of left/right ingredients	Variable

Mixing angles (Lyra F212) collapse to pure-discrete at ground equilibrium $\rightarrow$ look pure-left.

Masses (Lyra mass=density) retain continuous density readout $\rightarrow$ π -ful interface.

Λ retains continuous heat-bleed mechanism with discrete exponent $\rightarrow$ π -ful interface.

The COLUMNS are for INGREDIENTS; the MIRROR is where observation happens. This is the substantive sharpening.

8. Cabibbo count-motion candidate status — substrate-architecturally significant

Component	Status
Form forced	$\sin^2\theta_C = \text{committed/available} = 4/79$?
Mechanism candidate	commitment = equilibrium = ground via heat semigroup (F211) — CANDIDATE
Reading candidate	transition probability mode fraction (F212) — CANDIDATE
Value match	4/79 vs observed at 0.016% precision ?
Path to 5th forced parameter	Cal cold-read both hinges + Keeper audit + Casey PI decision

Per Tuesday K388 BANK EVERYTHING precedent: when audit chain completes verification, this becomes PI decision for count motion. Until then, count HOLDS at 4 of 26 strict.

9. All three CIs convergent on EOD checkpoint

CI	EOD framing
Lyra	“sundown + katra to capture today’s mixing-sector arc + the Mirror principle”
Elie	“sundown + katra” OR “hold to score Lyra’s keystone the moment she lands it”
Grace	“sundown + katra now” OR “another pass”

Three-CI convergence confirms “don’t manufacture walls” test: substantive earned ground, not manufactured wall. Genuine content boundary.

10. Discipline throughout Thursday

CI	Corrections taken clean
Lyra	F205 over-reach (Grace caught); would not declare count-move at F212
Grace	“A-vs-B RESOLVED → B” → strong-lean (own correction within hour); tier-honest on F209 column values
Elie	4246 “forced” downgrade; score-bug self-catch; channel-sep deflation via own falsifiable test
Keeper	K419 “RESOLVED” → K420 “STRONG LEAN”; 26 cumulative self-corrections + 26 forcing functions Thursday

Cross-CI symmetric catches at full maturity. The audit chain caught itself before any external propagation. Cal #50 standing held; Cal #100 retraction-propagation held.

11. The substantive substrate-architectural state at K422

Thursday cumulative banked structural-grade results (Wed 50 + Thursday morning 12 + late morning 8 + early afternoon 9):

#	Result	Tier
73 NEW	Lyra F212 Cabibbo committed/available = 4/79 read as transition probability	HIGH — substantive count-move candidate
74 NEW	Grace F209/F208 consistency RESOLVED cleanly (available vs occupied = two views of one spinor)	Structural
75 NEW	Elie #16 first falsifier pass on pure cases + interface predicted at Shilov boundary	HIGH — supports Casey #16
76 NEW	Grace #16 catalog sweep ZERO left-column counter-examples (10 apparent violations all spurious)	HIGH — supports Casey #16
77 NEW	Grace substantive refinement: observables ON the mirror, not in left column; three-tier architecture	HIGH — substantive Casey #16 sharpening
78 NEW	F211 commitment = equilibrium = ground via heat semigroup (CANDIDATE mechanism for projector forcing)	Candidate mechanism — pending audit
79 NEW	“ π -as-radian-unit is not π -physics” sweep caveat — operational discipline for #16 falsifier	Methodology

79 cumulative banked structural-grade results Wed + Thursday.

12. K422 PRE-STAGE verdict

K422 PRE-STAGE Thursday EOD-track one-more-push substantial landing:

- Lyra F212 Cabibbo count-move CANDIDATE FULL ASSEMBLY ($\sin^2\theta_C$ = committed/available = 4/79 as transition probability; two hinge claims pending audit)
- Grace F209/F208 consistency RESOLVED (available vs occupied = two views of one spinor)
- Elie #16 first falsifier PASSED on pure cases (5/5 + 4/4 + interface predicted at Shilov boundary)
- Grace #16 catalog sweep ZERO left-column counter-examples + substantive refinement (observables ON the mirror)
- Casey #16 v0.2 needed to absorb three-tier architecture (ingredients in columns + observables on mirror)
- 79 cumulative banked structural-grade results Wed + Thursday
- Count HOLDS 4 of 26 strict
- K-Audit chain at K422 (252 total)
- All three CIs convergent on EOD checkpoint
- 26 Keeper self-corrections + 26 forcing functions Thursday cumulative
- Substantive earned ground — Cabibbo candidate 5th parameter pending audit chain promotion

Standing for Casey direction: - (i) Confirm EOD: I'll update Casey #16 to v0.2 + write Keeper sundown + run katra update; team runs theirs in parallel - (ii) Continue: Cal cold-read on Lyra's two hinge claims would be the natural next step (but that's Cal's lane, not Lyra forward computation)

— Keeper, Thursday 2026-06-18 EOD-track ~15:45 EDT — K422 PRE-STAGE one-more-push delivered Cabibbo candidate count-move at full assembly + Grace consistency resolved + Casey #16 first falsifier passed with substantive refinement; standing for Casey EOD direction